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Studierichting: Informatica
Oefeningenreeks 1D nr. 9.1

Gegeven: $z = 1 + \sqrt{2}$

$z^2 = (1 + \sqrt{2})^2 = 3 + 2\sqrt{2}$

$z^3 = z^2 \cdot z = (3 + 2\sqrt{2})(1 + \sqrt{2}) = 7 + 5\sqrt{2}$

$z^4 = (z^2)^2 = (3 + 2\sqrt{2})^2 = 17 + 12\sqrt{2}$

$$\begin{aligned} z^4 - z^3 - 5z^2 + 4z + 2 &= (17 + 12\sqrt{2}) - (7 + 5\sqrt{2}) - 5(3 + 2\sqrt{2}) + 4(1 + \sqrt{2}) + 2 \\ &= (17 - 7 - 15 + 4 + 2) + (12 - 5 - 10 + 4)\sqrt{2} \\ &= 1 + \sqrt{2} \end{aligned}$$

References